DAY 05 ASSIGNMENT

ETHERNET AND MAC ADDRESS LEARNING

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Assignment Code: DAY05-ETHERNET-MAC

Topic: Ethernet and MAC Address Learning

1) THEORY

1. What is Ethernet?

Ethernet is widely used networking technology that is used for communication between devices within a Local Area Network (LAN). It is mainly used in wired networks to transfer data between computers, switches and routers. Ethernet has set standards for transmitting data across a network.

2. What is an Ethernet frame?

An Ethernet frame is a data unit that can be transferred between devices in an Ethernet network. It contains key pieces of information including the source and destination MAC addresses, data payload and error checking. Ethernet frames allow switches to send data to the correct devices within the local network.

3. What is a MAC address?

A MAC address is an address that relates to a network interface. It is used to uniquely identify network interfaces within a local network. Typically, Ethernet MAC addresses are 48 bits and usually written in hexadecimal format.

4. What does MAC stand for?

MAC stands for Media Access Control. MAC is a sublayer of the Data Link Layer in the OSI Model. It controls how devices communicate over a shared medium.

5. What is the difference between an MAC address and IP address?

A MAC address is used to uniquely identify a network interface at the Data Link Layer whereas an IP address is used to identify and locate devices within an IP network at the Network Layer.

A MAC address is mainly used within local networks, whereas an IP address helps data move between different networks. MAC addresses are usually associated with network interfaces whereas IP addresses can vary depending on the network.

6. What is a MAC Address Table?

A MAC Address Table is a table that is stored on a network switch that stores MAC addresses and the corresponding switch port. This helps the switch know which port has the MAC address of that device. This is also commonly referred to as a MAC table or CAM table.

7. How do switches learn MAC addresses?

Switches look at the source MAC address in the Ethernet frame. The source MAC address is stored along with the port it was received from and the relevant VLAN. This information is then stored in the switches MAC Address Table to help identify future frames.

8. What is a source MAC address?

A source MAC address is the MAC address of the device that sent an Ethernet frame. It helps identify which device has sent the frame on a local network. Switches use source mac addresses to learn which devices are connected to their ports.

9. What is a destination MAC address?

A destination MAC address is the MAC address of the network interface that the Ethernet frame is intended for. The switch will look up the destination MAC address in its MAC Address Table to know where to send the frame.

10. What is an unknown unicast frame?

An unknown unicast frame is a frame that the switch does not have a record of the destination MAC address in its MAC Address Table. As the switch does not know which port to send the frame to, the frame is flooded to all other eligible ports in the same VLAN.

11. What is flooding?

Flooding is when a switch sends a frame to multiple eligible ports within the same VLAN. This is mainly used when the destination MAC address is unknown to the switch.

12. What is known unicast forwarding?

Known unicast forwarding is when a switch receives a frame with a destination MAC address that is already in its MAC Address Table. The switch will only send the frame out of the port related to that specific MAC address. This makes communication more efficient.

13. What layer of the OSI Model does Ethernet Switching work on?

Ethernet switching works mainly on layer 2 of the OSI model. The Data Link Layer. Switches use MAC addresses to forward Ethernet frames.

14. Why would a switch have a MAC Address Table?

Switches have a MAC Address Table to identify which network devices are connected to which switch ports. This allows switches to send data to the correct destinations instead of sending to all devices. This makes networks more efficient.

15. Why is knowing about Ethernet important in Cyber Security?

It is important to know about Ethernet to understand how devices communicate within a local network. Knowing about MAC addresses, Ethernet frames and how switches work will help identify any unauthorised devices, investigate any suspicious network activity and understand network attacks.

Knowing about Ethernet will also help understand network attacks such as MAC spoofing and MAC flooding. This knowledge will allow Cyber Security professionals to monitor network communication and improve network device security.

2) PRACTICAL IDENTIFICATION

I used Windows PowerShell to identify my network adapter and view my MAC address and network information.

Network Adapter Name: Wireless LAN adapter Wi-Fi 2

MAC Address: 48-51-C5-A7-F6-66

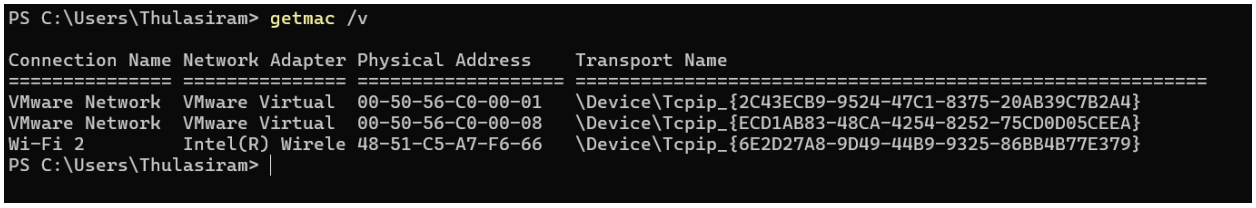
IPv4 Address: 10.47.230.97

Default Gateway: 10.47.230.60

3) SCREENSHOTS

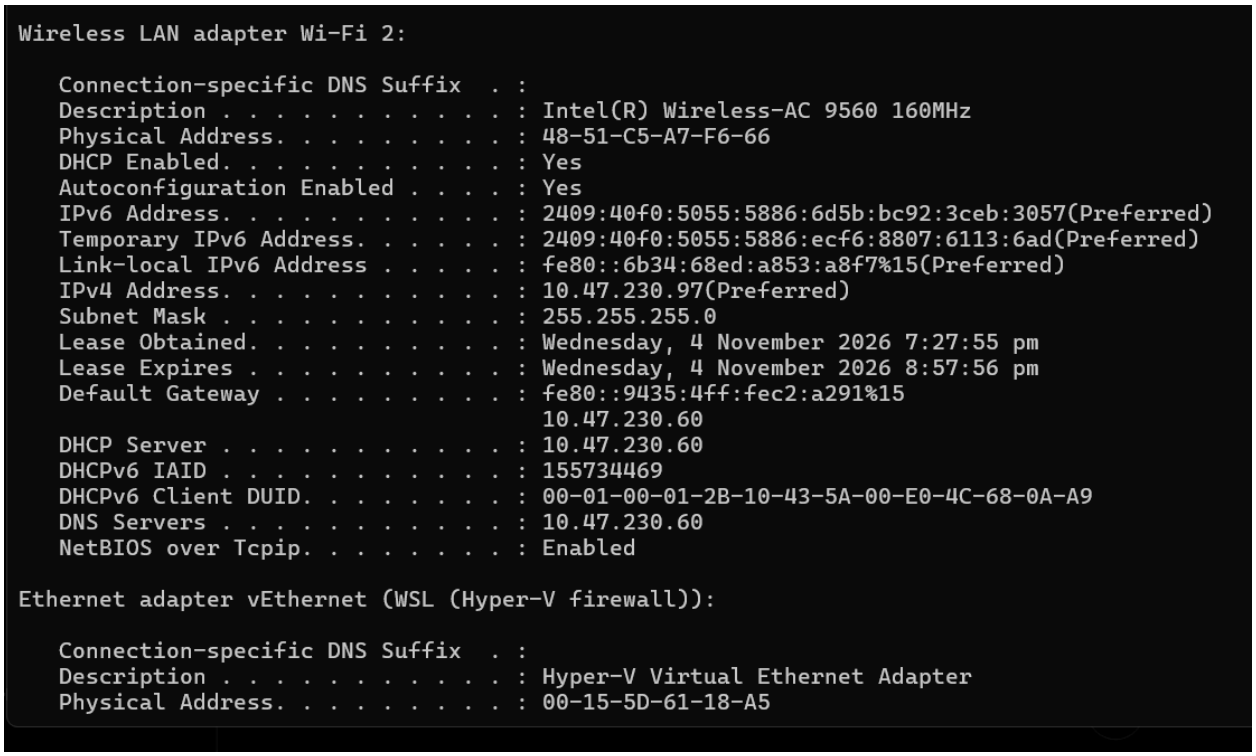
Screenshot 1 – GETMAC Command

This screenshot shows the results of using the 'getmac /v' command. It shows information about the network connection, network adapter and physical MAC address.



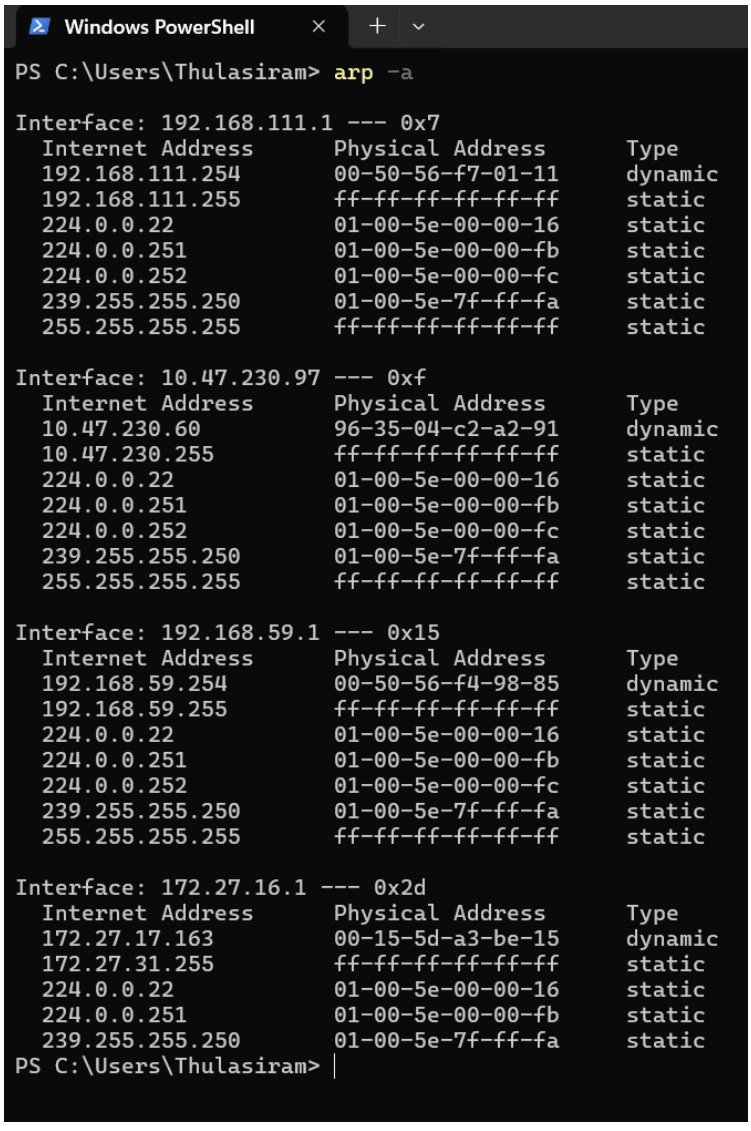
Screenshot 2 – IP Configuration

This screenshot shows the results of using the 'ipconfig /all' command. It shows information about the active network adapter, MAC address, private IPv4 address and default gateway.



Screenshot 3 – ARP Table

This screenshot shows the results of using the 'arp -a' command. It shows the IP addresses and MAC addresses that are stored in my computers ARP cache.



CONCLUSION

By completing this task I have learnt about Ethernet, Ethernet frames, MAC Addresses and how switches use MAC Address Tables to send data. I understand how switches learn MAC

addresses, the difference between unknown unicast flooding and known unicast forwarding. I have learnt how to use the ‘getmac’, ‘ipconfig/all’ and ‘arp -a’ commands to identify my network adapter, MAC address and network information. This knowledge can be useful to understand how devices communicate within networks and how Cyber Security professionals identify and investigate suspicious network activity.